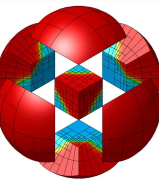




中山大學
SUN YAT-SEN UNIVERSITY

MFEM Community
Workshop, 25/09/2026



A Two-scale Higher-Order Damage Elasticity Theory and Solution Procedure for Quasi-brittle Fracture

Zhang Chunyu, Cao Yuheng

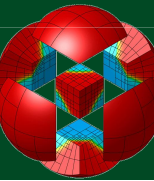
Sino-French Institute of Nuclear Engineering & Technology

Sun Yat-Sen University

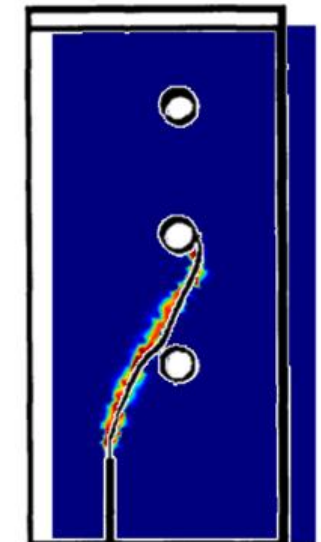
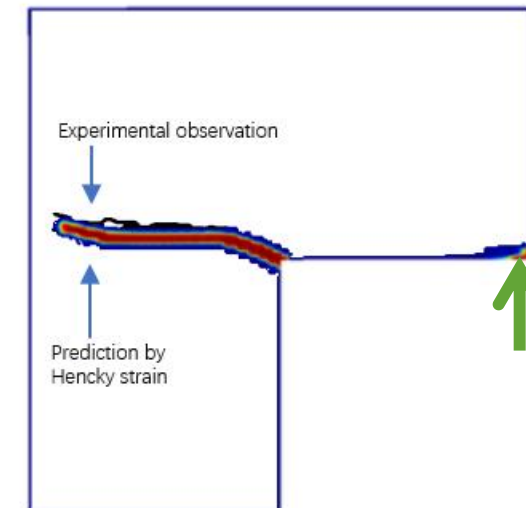
Email: zhangchy5@mail.sysu.edu.cn

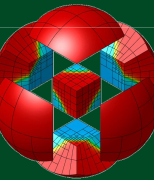
25/09/2026



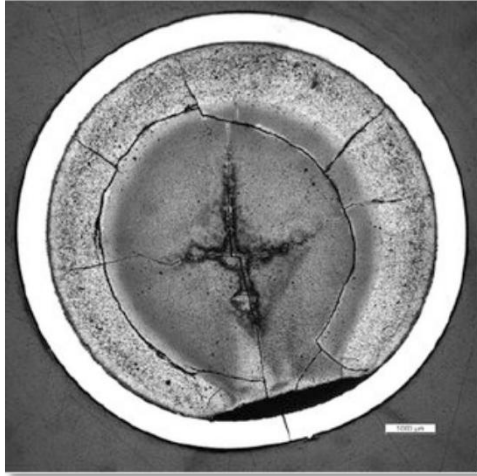


1. From damage modelling to fracture prediction: the gap
2. Proposed remedy: resolve sub-RVE inhomogeneity at the macro-scale
3. Solution method and FE implementation
4. Results
5. Conclusion and future work

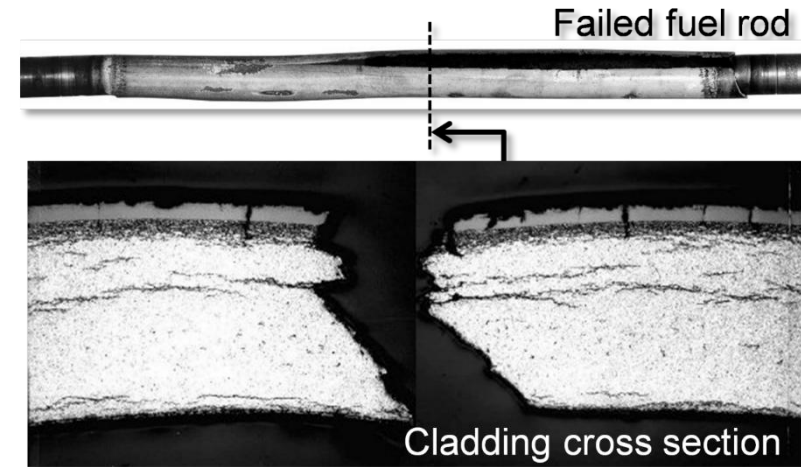




- Structural integrity assessment: nuclear fuel failure

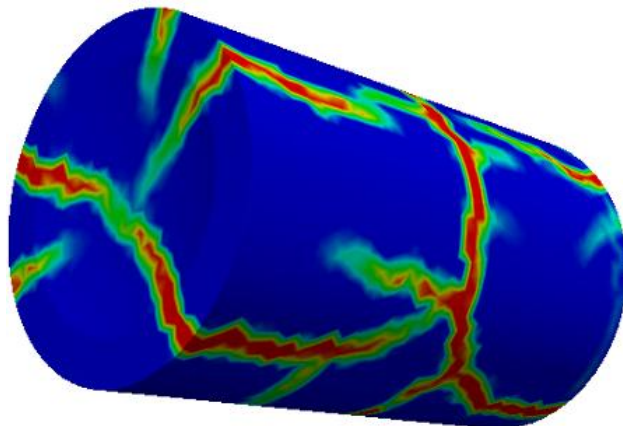


Capps et al., 2015

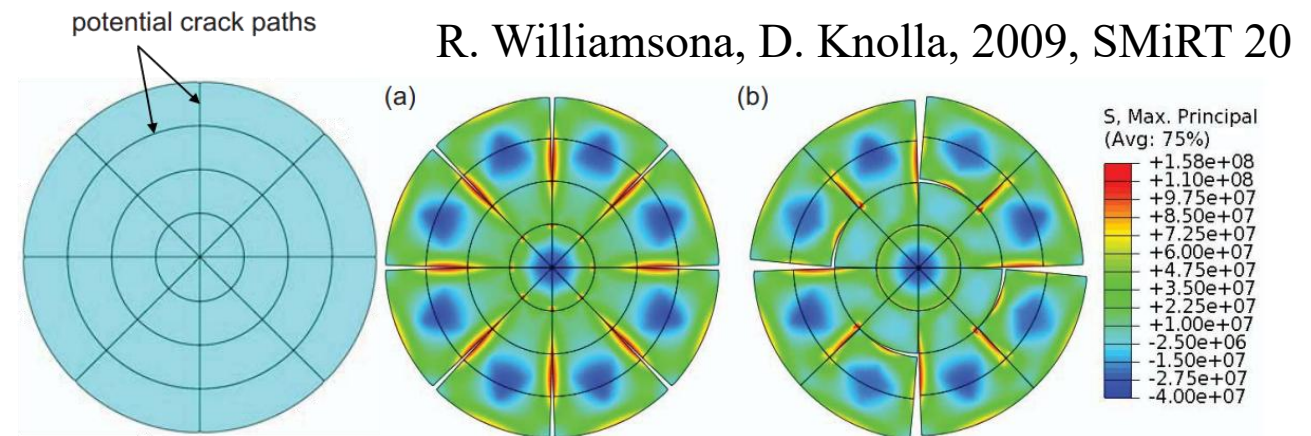


JAEA

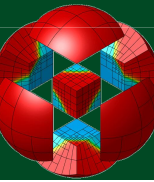
- Damage mechanics & fracture mechanics: two approaches to a coupled process



Fuel pellet
fragmentation.
Cast3M
B. Michel
2014



R. Williamsona, D. Knolla, 2009, SMiRT 20

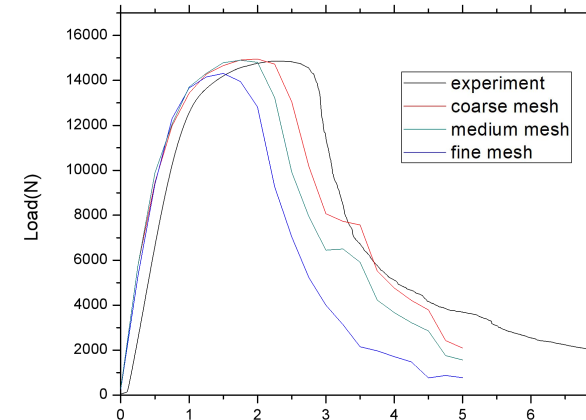


Problem

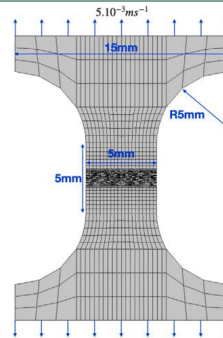
Mesh size sensitivity



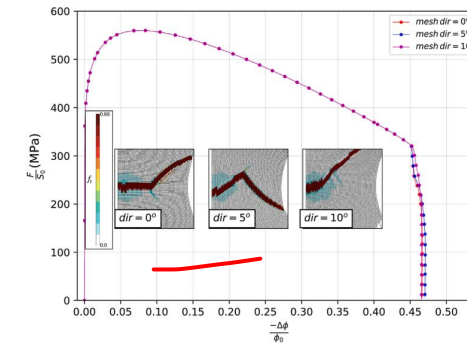
GTN
model



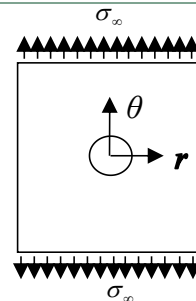
Mesh layout dependence (Mech. Mater. 171 (2022) 104350)



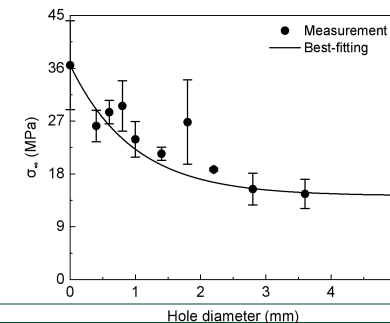
Non-local
GTN model

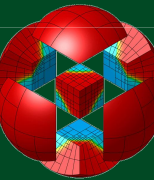


Size effect (Mech. Mater. 187 (2023) 104841)

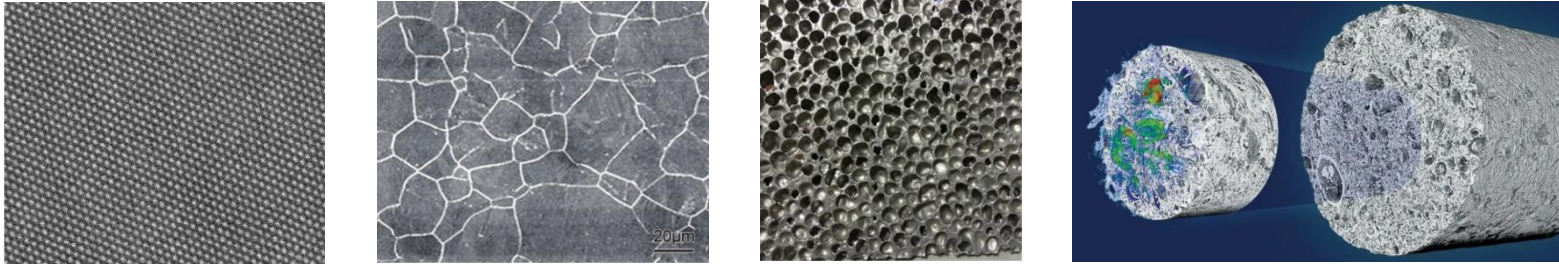


PMMA

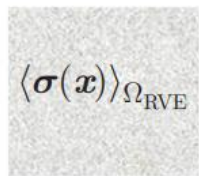




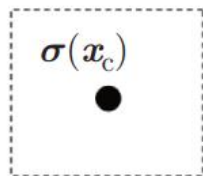
Real material: multi-scale and heterogeneous



Continuum model: homogenized and averaged

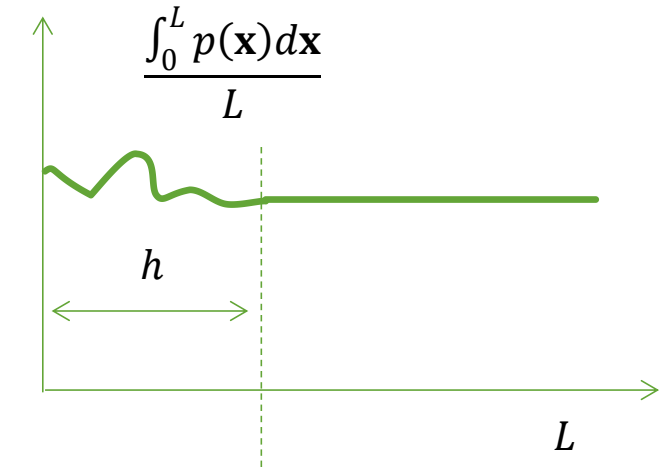


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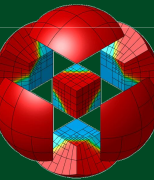
$$\langle p \rangle_{\Omega_{RVE}} := \frac{\int_{\Omega_{RVE}} p(\mathbf{x}) d\mathbf{x}}{\int_{\Omega_{RVE}} d\mathbf{x}} \quad \text{Filtered field}$$

$p(\mathbf{x}_c)$: Field defined at the centroid of RVE

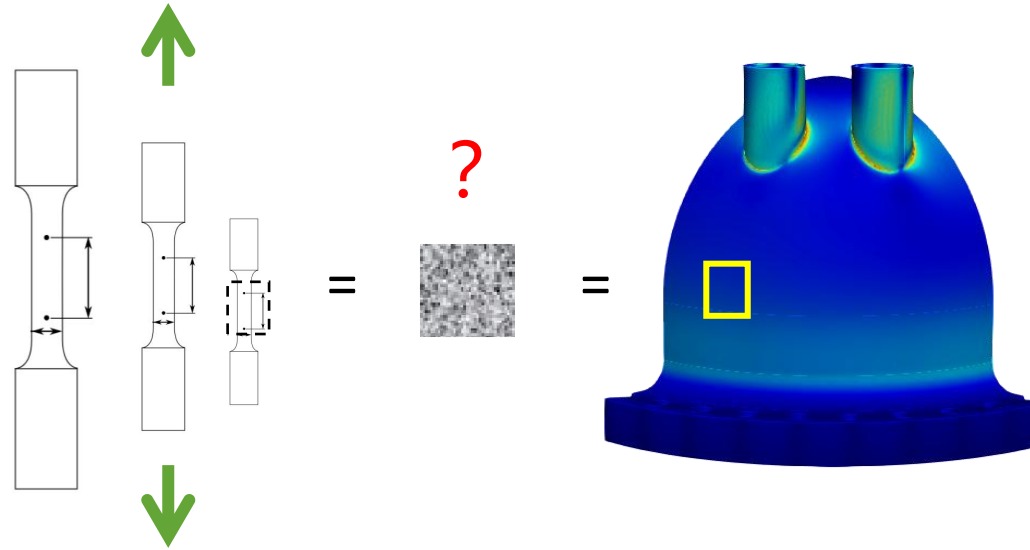


RVE: small but finite-sized

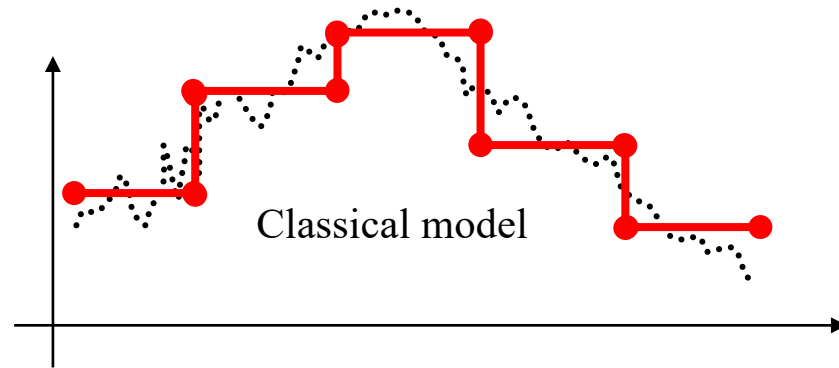
Material	RVE size
Polymer	0.5~1mm
Alloy	10~100μm
Concrete	1~5cm



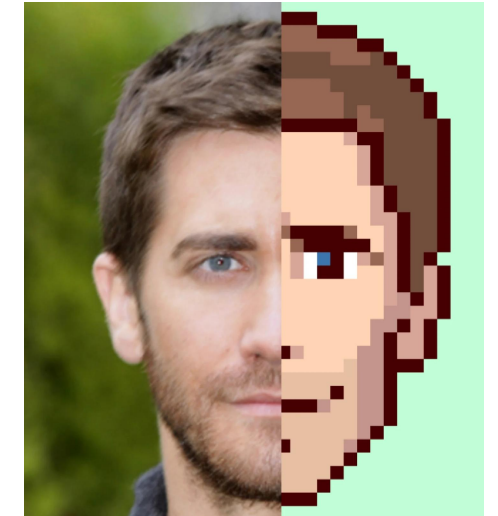
Constitutive model: inferred from **homogeneous** deformations



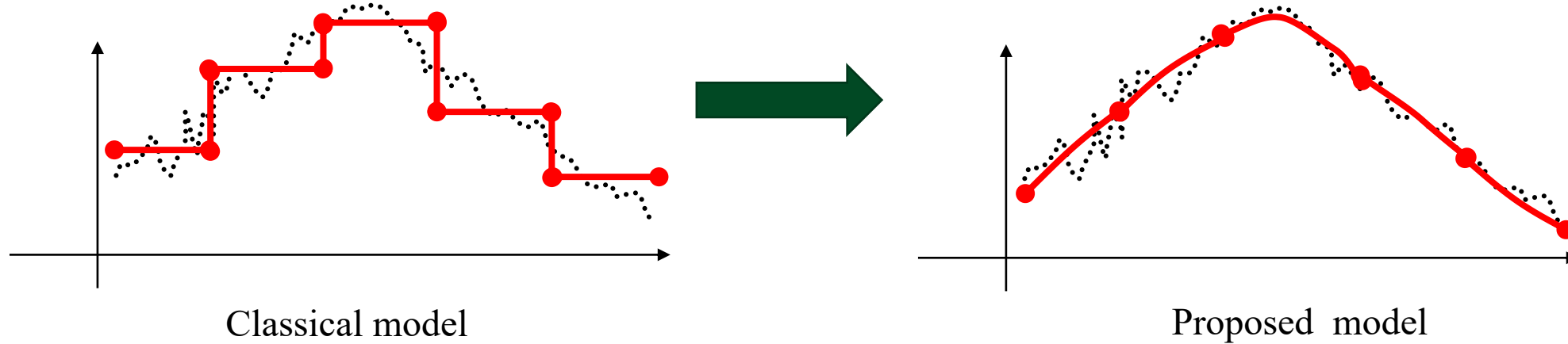
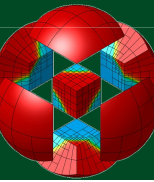
applied to **inhomogeneous** deformations



- Sub-RVE unresolved
- Pronounced modeling errors under highly inhomogeneous deformation (feature size $\approx$ RVE size)



Proposed remedy: resolve sub-RVE inhomogeneity at the macro-scale



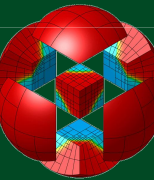
- Sub-RVE inhomogeneity recovered from macro-strain gradients
- Constitutive model defined by homogenized energy density incorporating sub-RVE strain
- Single length scale parameter with clear physical meaning

$$\bar{U} = \frac{1}{h^3} \iiint_{-\frac{h}{2}}^{\frac{h}{2}} \left(U + \nabla U \cdot \mathbf{x} + \frac{1}{2} (\nabla^2 U : \mathbf{x}) : \mathbf{x} + \dots \right) dx_1 dx_2 dx_3$$

$$= U(0) + \boxed{\frac{h^2}{24} \Delta U(0) + \frac{h^4}{1920} \Delta^2 U(0) + \frac{h^4}{2880} U_{,kkl(l(k \neq l))}(0) \dots}$$

Physics	U
Elasticity	Strain energy density
Quasi-brittle fracture	Degraded strain energy density
Plasticity	Plastic dissipation
Thermal	Specific enthalpy

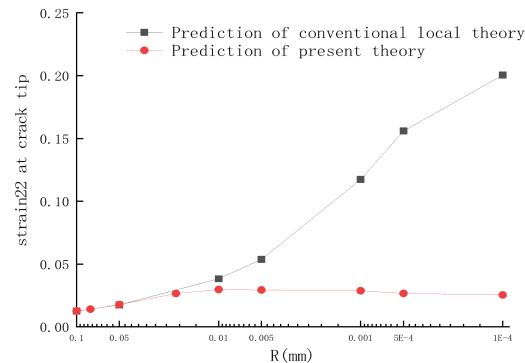
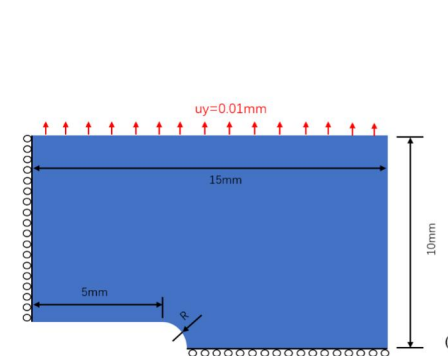
Example #1: higher-order elasticity



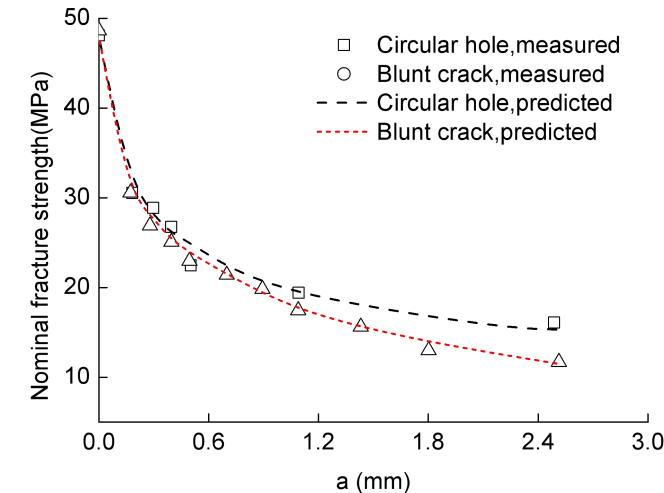
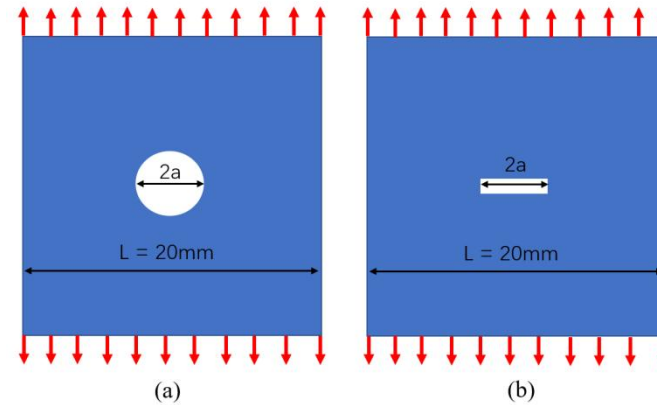
$$1. U(\boldsymbol{\varepsilon}) = \frac{\lambda}{2} \varepsilon_{ii} \varepsilon_{jj} + \mu \varepsilon_{ij}^2$$

$$2. \bar{U}(\boldsymbol{\varepsilon}) = U + \frac{h^2}{24} \Delta U + \frac{h^4}{1920} \Delta^2 U + \frac{h^4}{2880} U_{,kkll} (k \neq l)$$

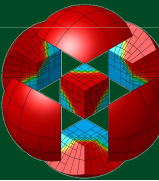
$$= \frac{\lambda}{2} \varepsilon_{ii} \varepsilon_{jj} + \mu \varepsilon_{ij}^2 + \frac{\lambda}{2} \left(\frac{h^2}{12} \left(\varepsilon_{ii} \varepsilon_{jj, kk} + \varepsilon_{ii, k} \varepsilon_{jj, k} \right) + \frac{h^4}{4} \left(\frac{1}{144} \varepsilon_{ii, kk} \varepsilon_{jj, ll} + \frac{1}{72} \varepsilon_{ii, kl} \varepsilon_{jj, kl} - \frac{1}{120} \varepsilon_{ii, kk} \varepsilon_{jj, kk} \right) \right) + \mu \left(\frac{h^2}{12} \left(\varepsilon_{ij, k}^2 + \varepsilon_{ij} \varepsilon_{ij, kk} \right) + \frac{h^4}{4} \left(\frac{1}{144} \varepsilon_{ij, kk} \varepsilon_{ij, ll} + \frac{1}{72} \varepsilon_{ij, kl} \varepsilon_{ij, kl} - \frac{1}{120} \varepsilon_{ij, kk} \varepsilon_{ij, kk} \right) \right)$$



Retain h^4 terms → Eliminate crack-tip deformation singularity



Example #2: higher-order damage elasticity



$$1. U_d(\underline{\epsilon}) = (1 - d)U = (1 - d) \left(\frac{\lambda}{2} \epsilon_{ii} \epsilon_{jj} + \mu \epsilon_{ij}^2 \right)$$

$$2. \widetilde{U}_d = (1 - d)U + \frac{h^2}{24} \Delta \{ (1 - d)U \} + \frac{h^4}{4} \left(\frac{\{ (1-d)U \}_{kkll(k \neq l)}}{288} + \frac{\{ (1-d)U \}_{kkkk}}{480} \right)$$

$$3. \widetilde{U}_d = (1 - d)\widetilde{U} - \widetilde{U}_d^{h,2} - \widetilde{U}_d^{h,4}$$

$$= (1 - d)\widetilde{U} - \frac{h^2}{24} \left(\frac{\lambda}{2} \underline{\underline{I}}_d + \mu \underline{\underline{1}} \right) : \left(\Delta d \cdot \underline{\underline{\epsilon}} \underline{\underline{\epsilon}}^T + 4 \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}} \overline{\nabla} d)^T \right)$$

$$- \frac{h^4}{4} \left(\frac{\lambda}{2} \underline{\underline{I}}_d + \mu \underline{\underline{1}} \right) : \left(\frac{1}{36} \left(2 \overline{\nabla} \underline{\underline{\epsilon}} (\overline{\nabla}^2 \underline{\underline{\epsilon}} : \overline{\nabla} d)^T + \Delta \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}} \overline{\nabla} d)^T \right) + \frac{1}{36} \left(\overline{\nabla} \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}} \overline{\nabla}^2 d)^T + \underline{\underline{\epsilon}} (\overline{\nabla}^2 \underline{\underline{\epsilon}} : \overline{\nabla}^2 d)^T \right) \right.$$

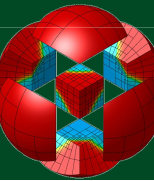
$$\left. + \frac{1}{72} \left(\Delta d \cdot \overline{\nabla} \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}})^T + \Delta d \cdot \Delta \underline{\underline{\epsilon}} \underline{\underline{\epsilon}}^T \right) - \frac{1}{30} \overline{\nabla} d \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}} \underline{\underline{d}}_{xx})^T - \frac{1}{60} \left(\overline{\nabla} \underline{\underline{\epsilon}} (\overline{\nabla} \underline{\underline{\epsilon}} \underline{\underline{d}}_{xx})^T + \underline{\underline{\epsilon}} (\overline{\nabla} d \underline{\underline{\epsilon}} \overline{\nabla} d)^T \right) \right)$$

with

$$\underline{\underline{\epsilon}} = \{\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{12}, \epsilon_{13}, \epsilon_{23}, \epsilon_{21}, \epsilon_{31}, \epsilon_{32}\}^T$$

$$\underline{\underline{I}}_d = \{1, 1, 1, 0, 0, 0, 0, 0, 0\} \in \mathbb{R}^9$$

- Complex mathematics but simple physics
- Retain h^4 terms $\rightarrow$ Robust results



Idea: damage evolution is driven exclusively by the **tensile** part of the strain energy.

Spectral decomposition of strain (Miehe et al., Int. J. Numer. Methods Eng. 83 (2010) 1273–1311):

$$\underline{\underline{\boldsymbol{\varepsilon}}}^\pm = \sum_{i=1}^3 \langle \varepsilon_i \rangle_\pm \underline{\underline{\mathbf{n}}}_i \otimes \underline{\underline{\mathbf{n}}}_i$$

$$U = U^+ + U^-, U^\pm = \frac{\lambda}{2} \left(\left\langle \text{tr} \left(\underline{\underline{\boldsymbol{\varepsilon}}} \right) \right\rangle_\pm \right)^2 + \mu \text{tr} \left(\left(\underline{\underline{\boldsymbol{\varepsilon}}}^\pm \right)^2 \right)$$

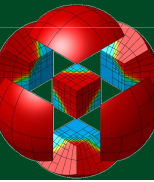
Explicit local form:

$$U^\pm = \underline{\underline{\boldsymbol{\varepsilon}}} : \underline{\underline{\mathbf{D}}}^\pm : \underline{\underline{\boldsymbol{\varepsilon}}} \quad U_d = (1 - d) \underline{\underline{\boldsymbol{\varepsilon}}} : \underline{\underline{\mathbf{D}}}^+ : \underline{\underline{\boldsymbol{\varepsilon}}} + \underline{\underline{\boldsymbol{\varepsilon}}} : \underline{\underline{\mathbf{D}}}^- : \underline{\underline{\boldsymbol{\varepsilon}}}$$

$$\underline{\underline{\mathbf{D}}}^\pm \equiv \frac{\lambda}{2} \left(H_v \left(\text{tr} \left(\underline{\underline{\boldsymbol{\varepsilon}}} \right) \right) + H_v \left(-\text{tr} \left(\underline{\underline{\boldsymbol{\varepsilon}}} \right) \right) \right) \underline{\underline{\mathbf{I}}} + \mu \sum_{i=1}^3 H_v(\pm \varepsilon_i) \underline{\underline{\mathbf{n}}}^i \otimes \underline{\underline{\mathbf{n}}}^i \otimes \underline{\underline{\mathbf{n}}}^i \otimes \underline{\underline{\mathbf{n}}}^i + \mu \sum_{i=1}^3 \sum_{j \neq i}^3 \frac{\langle \varepsilon_i \rangle_\pm - \langle \varepsilon_j \rangle_\pm}{\varepsilon_i - \varepsilon_j} \underline{\underline{\mathbf{n}}}^i \otimes \underline{\underline{\mathbf{n}}}^j \otimes \underline{\underline{\mathbf{n}}}^i \otimes \underline{\underline{\mathbf{n}}}^j$$

Non-local form:

$$\widetilde{U}_d^* = (1 - d) \widetilde{U}^+ - \widetilde{U}_{d+}^{\widetilde{h},2} - \widetilde{U}_{d+}^{\widetilde{h},4} + \widetilde{U}^-$$



Hencky strain tensor:

$$\underline{\underline{E}} = \frac{1}{2} \ln(\underline{\underline{C}}) = \frac{1}{2} \sum_{k=1}^3 \ln(c_k) \underline{\underline{n}}_k \otimes \underline{\underline{n}}_k \quad \underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}} \quad \underline{\underline{F}} = \frac{\partial \underline{\underline{x}}}{\partial \underline{\underline{X}}} := \overline{\underline{\underline{V}}} \underline{\underline{x}}$$

Tension–compression decomposition(M. Ortiz, 1985):

$$\underline{\underline{E}}^\pm = \sum_{k=1}^3 \langle \ln(c_k) \rangle_\pm \underline{\underline{n}}_k \otimes \underline{\underline{n}}_k$$

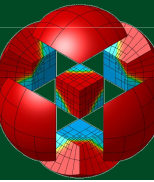
Local elastic strain energy density:

$$U = U^+ + U^-, U^\pm = \frac{\lambda}{2} \left(\left\langle \text{tr}(\underline{\underline{E}}) \right\rangle_\pm \right)^2 + \mu \text{tr} \left((\underline{\underline{E}}^\pm)^2 \right)$$

Explicit local form:

$$U_d = (1 - d) \underline{\underline{E}} : \underline{\underline{D}}^+ : \underline{\underline{E}} + \underline{\underline{E}} : \underline{\underline{D}}^- : \underline{\underline{E}}$$

Solution method: minimum of potential energy with damage evolution



$$\begin{cases} J(\underline{\mathbf{u}}, d) = \int_{\Omega} \widetilde{U}_d(\underline{\boldsymbol{\varepsilon}}(\underline{\mathbf{u}}), d) dV - \int_{\partial\Omega} \underline{\mathbf{u}} \cdot \underline{\mathbf{t}} dS \\ d - f(\overline{\boldsymbol{\varepsilon}}, t) = 0 \end{cases}$$

$$\frac{\delta J}{\delta \underline{\mathbf{u}}} = 0 \Leftrightarrow \int_{\Omega} \left[(1-d) \delta \underline{\mathbf{u}} \widetilde{U} - \delta \underline{\mathbf{u}} \widetilde{U}_d^{h,2} - \delta \underline{\mathbf{u}} \widetilde{U}_d^{h,4} \right] dV - \int_{\partial\Omega} \underline{\mathbf{t}}^T \delta \underline{\mathbf{u}} dS = 0$$

$$d = \begin{cases} 0, \overline{\boldsymbol{\varepsilon}}_t^+ < \overline{\boldsymbol{\varepsilon}}_c \\ 1 - \left\{ \alpha + (1-\alpha) \frac{\overline{\boldsymbol{\varepsilon}}_c}{\overline{\boldsymbol{\varepsilon}}_t^+} \exp\left(-\beta \left(\overline{\boldsymbol{\varepsilon}}_t^+ - \overline{\boldsymbol{\varepsilon}}_c\right)\right) \right\}, \overline{\boldsymbol{\varepsilon}}_t^+ \geq \overline{\boldsymbol{\varepsilon}}_c \end{cases}$$

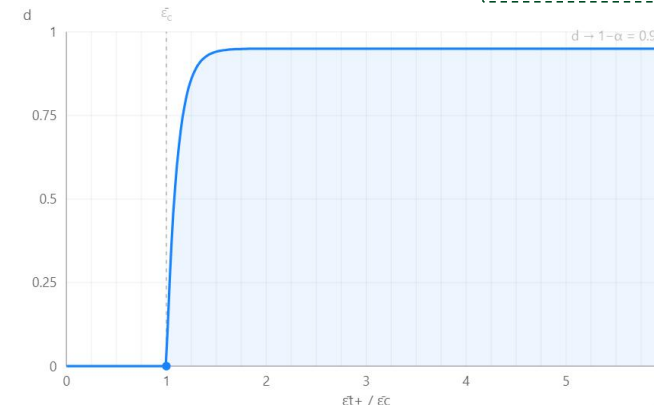
$$\overline{\boldsymbol{\varepsilon}}^+ = \sqrt{\frac{2U^+}{E}} = \sqrt{\frac{\underline{\boldsymbol{\varepsilon}}^T \underline{\mathbf{D}}^+ \underline{\boldsymbol{\varepsilon}}}{E}}$$

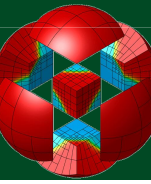
$$\overline{\boldsymbol{\varepsilon}}_t(t) = \max\{\overline{\boldsymbol{\varepsilon}}(\tau), \tau \in \llbracket 0, t \rrbracket\}$$

$\alpha = 10^{-9}$: ensure numerical stability during the final stages of failure

β : material toughness controlling the softening response

$\overline{\boldsymbol{\varepsilon}}_c$: critical equivalent strain for crack initiation





$$\frac{\delta J}{\delta \underline{\mathbf{u}}} = 0 \Leftrightarrow \int_{\Omega} \left[(1-d) \delta \underline{\mathbf{u}} \tilde{U} - \delta \underline{\mathbf{u}} \widetilde{U_d^{h,2}} - \delta \underline{\mathbf{u}} \widetilde{U_d^{h,4}} \right] dV - \int_{\partial\Omega} \underline{\mathbf{t}}^T \delta \underline{\mathbf{u}} dS = 0$$

↓

1. Challenge: gradient orders ≥ 3
2. Standard method: integration by parts $\rightarrow$ higher-order gradients to BCs
3. Our method: strain $\underline{\boldsymbol{\varepsilon}}$ as an independent field and reformulate total potential energy by ALM

$$\begin{aligned} J_{ALM}(\underline{\mathbf{u}}, \underline{\boldsymbol{\varepsilon}}, d, \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}) &= \int_{\Omega} \widetilde{U}_d(\underline{\boldsymbol{\varepsilon}}, d) dV - \int_{\partial\Omega} \underline{\mathbf{u}} \cdot \underline{\mathbf{t}} dS + \int_{\Omega} \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}^T (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}}) dV \\ &+ \int_{\Omega} \frac{r}{2} (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}})^T (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}}) dV \end{aligned}$$

$\underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}$: Lagrangian multiplier
 r : penalty parameter

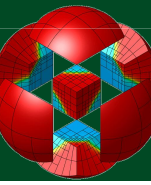
4. The kinematic constraint in finite deformation

$$\underline{\underline{\mathbf{E}}} = \frac{1}{2} \ln(\underline{\underline{\mathbf{F}}}^T \underline{\underline{\mathbf{F}}})$$

Cao Y.H., Zhang C.Y., CMAME 445(2025), 118206.

Cao Y.H., Zhang C.Y., Eng. Frac. Mech., 2026, submitted

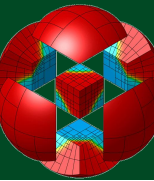
Solution method: equivalent lower-order variational formulation



$$J_{ALM}(\underline{\mathbf{u}}, \underline{\boldsymbol{\varepsilon}}, d, \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}) = \int_{\Omega} \widetilde{U}_d(\underline{\boldsymbol{\varepsilon}}, d) dV - \int_{\partial\Omega} \underline{\mathbf{u}} \cdot \underline{\mathbf{t}} dS + \int_{\Omega} \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}^T (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}}) dV + \int_{\Omega} \frac{r}{2} (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}})^T (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}}) dV$$

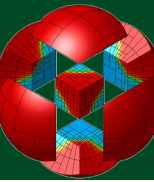
$$\left\{ \begin{array}{l} \textcircled{1} \frac{\delta J_{ALM}^d}{\delta \underline{\mathbf{u}}} = 0 \Leftrightarrow \int_{\Omega} \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}^T \delta \underline{\nabla}_s \underline{\mathbf{u}} dV + r \int_{\Omega} (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}})^T \delta \underline{\nabla}_s \underline{\mathbf{u}} dV - \int_{\partial\Omega} \underline{\mathbf{t}}^T \delta \underline{\mathbf{u}} dS = 0 \\ \textcircled{2} \frac{\delta J_{ALM}^d}{\delta \underline{\boldsymbol{\varepsilon}}} = 0 \Leftrightarrow \int_{\Omega} (1-d) \delta \underline{\boldsymbol{\varepsilon}} \widetilde{U} - \delta \underline{\boldsymbol{\varepsilon}} \widetilde{U}_d^{h,2} - \delta \underline{\boldsymbol{\varepsilon}} \widetilde{U}_d^{h,4} dV + r \int_{\Omega} (\underline{\boldsymbol{\varepsilon}} - \underline{\nabla}_s \underline{\mathbf{u}})^T \delta \underline{\boldsymbol{\varepsilon}} dV - \int_{\Omega} \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}^T \delta \underline{\boldsymbol{\varepsilon}} dV = 0 \\ \textcircled{3} \frac{\delta J_{ALM}^d}{\delta \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}} = 0 \Leftrightarrow \int_{\Omega} (\underline{\nabla}_s \underline{\mathbf{u}} - \underline{\boldsymbol{\varepsilon}})^T \delta \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}} dV = 0 \\ \textcircled{4} d = 1 - \left\{ \alpha + (1-\alpha) \frac{\overline{\varepsilon}_c}{\varepsilon_t^+} \exp\left(-\beta (\overline{\varepsilon}_t^+ - \overline{\varepsilon}_c)\right) \right\}, \overline{\varepsilon}_t^+ \geq \overline{\varepsilon}_c \end{array} \right.$$

Gradient orders ≤ 2 for all fields: $\underline{\mathbf{u}}, \underline{\boldsymbol{\varepsilon}}, \underline{\boldsymbol{\lambda}}_{\boldsymbol{\varepsilon}}$



$$\left\{ \begin{array}{l} \textcircled{1} \frac{\delta J_{ALM}^d}{\delta \underline{\underline{\mathbf{u}}}} = 0 \Leftrightarrow \int_{\Omega} \underline{\underline{\lambda}}_{\underline{\underline{\epsilon}}}^T \delta \underline{\underline{\nabla}}_s \underline{\underline{\mathbf{u}}} dV + r \int_{\Omega} \left(\underline{\underline{\nabla}}_s \underline{\underline{\mathbf{u}}} - \underline{\underline{\epsilon}} \right)^T \delta \underline{\underline{\nabla}}_s \underline{\underline{\mathbf{u}}} dV - \int_{\partial\Omega} \underline{\underline{\mathbf{t}}}^T \delta \underline{\underline{\mathbf{u}}} dS = 0 \\ \textcircled{2} \frac{\delta J_{ALM}^d}{\delta \underline{\underline{\epsilon}}} = 0 \Leftrightarrow \int_{\Omega} (1-d) \delta \underline{\underline{\epsilon}} \tilde{U} - \delta \underline{\underline{\epsilon}} \tilde{U}_d^{\tilde{h},2} - \delta \underline{\underline{\epsilon}} \tilde{U}_d^{\tilde{h},4} dV + r \int_{\Omega} \left(\underline{\underline{\epsilon}} - \underline{\underline{\nabla}}_s \underline{\underline{\mathbf{u}}} \right)^T \delta \underline{\underline{\epsilon}} dV - \int_{\Omega} \underline{\underline{\lambda}}_{\underline{\underline{\epsilon}}}^T \delta \underline{\underline{\epsilon}} dV = 0 \\ \textcircled{3} \frac{\delta J_{ALM}^d}{\delta \underline{\underline{\lambda}}_{\underline{\underline{\epsilon}}}} = 0 \Leftrightarrow \int_{\Omega} \left(\underline{\underline{\nabla}}_s \underline{\underline{\mathbf{u}}} - \underline{\underline{\epsilon}} \right)^T \delta \underline{\underline{\lambda}}_{\underline{\underline{\epsilon}}} dV = 0 \\ \textcircled{4} d = 1 - \left\{ \alpha + (1-\alpha) \frac{\overline{\epsilon}_c}{\epsilon_t^+} \exp \left(-\beta \left(\overline{\epsilon}_t^+ - \overline{\epsilon}_c \right) \right) \right\}, \overline{\epsilon}_t^+ \geq \overline{\epsilon}_c \end{array} \right.$$

1. Given d , solve $\textcircled{1}\textcircled{2}\textcircled{3}$ to obtain $\underline{\underline{\mathbf{u}}}, \underline{\underline{\epsilon}}, \underline{\underline{\lambda}}_{\underline{\underline{\epsilon}}}$
2. Given $\underline{\underline{\epsilon}}$, update d via $\textcircled{4}$
3. Repeat steps 1-2 until convergence

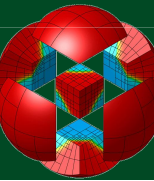


$$\begin{aligned} \underline{\underline{K_{\varepsilon\varepsilon}}}_{i,j} &:= \int_{\Omega} \left(2 \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}} \underline{\underline{\varphi_{\varepsilon}^j}} + \frac{h^2}{12} \left(\Delta \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}} \underline{\underline{\varphi_{\varepsilon}^j}} + \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}} \Delta \underline{\underline{\varphi_{\varepsilon}^j}} + 2 \bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}}_{\nabla} \bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \right) \right. \\ &\quad \left. + \frac{h^4}{4} \left(\frac{1}{72} \Delta \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}} \Delta \underline{\underline{\varphi_{\varepsilon}^j}} + \frac{1}{36} \left(\bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}}_{2\nabla} \bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^j}} \right) - \frac{1}{60} \bar{\nabla}_{dd} \underline{\underline{\varphi_{\varepsilon}^i}}^T \underline{\underline{D}}_{\nabla} \bar{\nabla}_{dd} \underline{\underline{\varphi_{\varepsilon}^j}} \right) \right) dV \end{aligned}$$

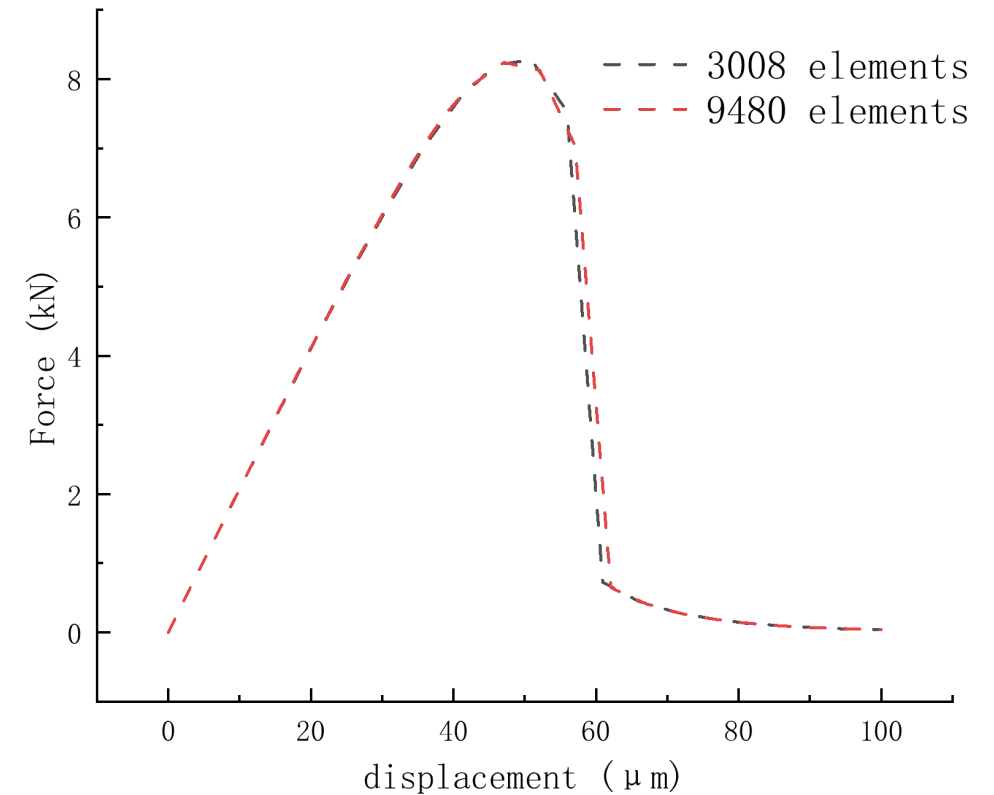
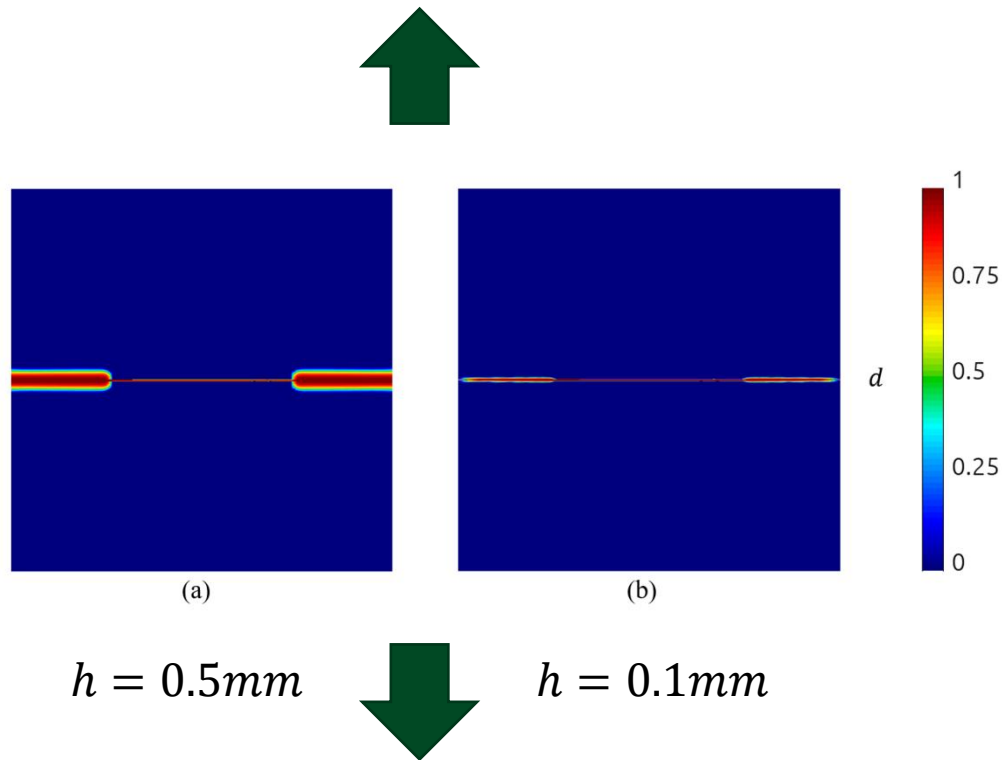
- C⁰ quadratic Lagrange elements
- Core functions: *CalcPhysHessian* and *CalcPhysLaplacian*
(<https://github.com/mfem/mfem/issues/3367>)
- *DamageIntegratorEE*(BilinearFormIntegrator for small strain formulation) and
DHEFiniteDeformNewtonIntegrator
(NonlinearFormIntegrator for finite deformation formulation)

$$\begin{aligned} \underline{\underline{K_{\varepsilon\varepsilon}^d}}_{i,j} &:= \int_{\Omega} (1-d) \underline{\underline{K_{\varepsilon\varepsilon}}}_{i,j} \\ &\quad - \frac{h^2}{12} \left(\Delta d \cdot \left(\underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}} \underline{\underline{\varphi_{\varepsilon}^j}} + 2 \left(\underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}} \left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^j + 2 \left(\left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^i \right)^T \underline{\underline{D}} \underline{\underline{\varphi_{\varepsilon}^j}} \right) \\ &\quad - \frac{h^4}{4} \left(\frac{1}{36} \left(2 \left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}} \left(\bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^j + 2 \left(\left(\bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^i \right)^T \underline{\underline{D}}_{\nabla} \bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \right. \right. \\ &\quad \left. \left. + \left(\Delta \underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}} \left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^j + \left(\left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla} d \right)^i \right)^T \underline{\underline{D}} \Delta \underline{\underline{\varphi_{\varepsilon}^j}} \right) \right. \\ &\quad \left. + \frac{1}{36} \left(\left(\left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla}^2 d \right)^i \right)^T \underline{\underline{D}}_{\nabla} \bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} + \left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}}_{\nabla} \left(\bar{\nabla} \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla}^2 d \right)^j + \left(\left(\bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla}^2 d \right)^i \right)^T \underline{\underline{D}} \underline{\underline{\varphi_{\varepsilon}^j}} \right. \right. \\ &\quad \left. \left. + \left(\underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{D}} \left(\bar{\nabla}^2 \underline{\underline{\varphi_{\varepsilon}^j}} \bar{\nabla}^2 d \right)^j \right) + \dots \right) + r \left(\underline{\underline{\varphi_{\varepsilon}^i}} \right)^T \underline{\underline{\varphi_{\varepsilon}^j}} dV \end{aligned}$$

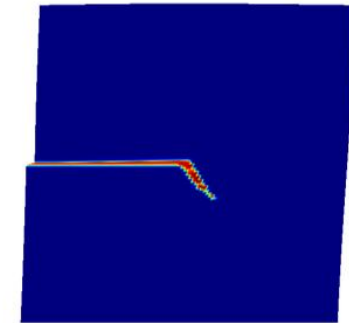
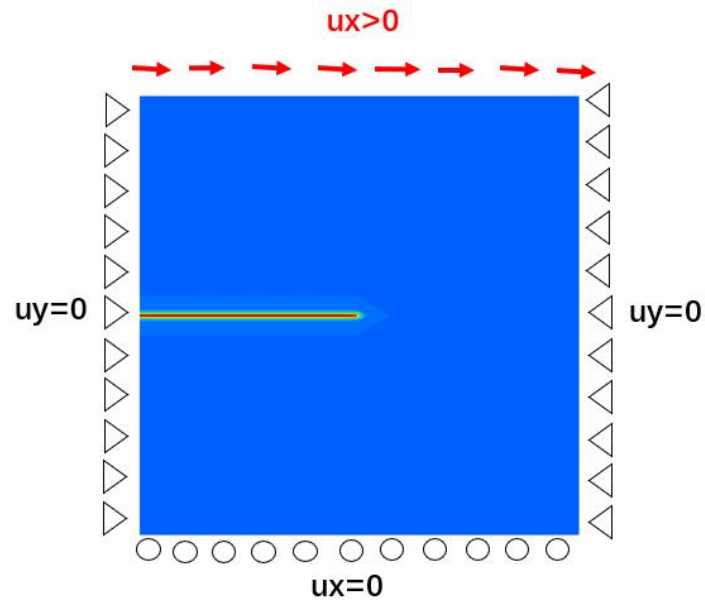
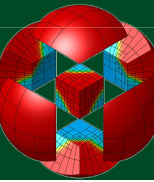
Cao Y.H., Zhang C.Y., Wang B., Int. J. Eng. Sci. 195 (2024) 103990.
Cao Y.H., Zhang C.Y., Comput. Methods Appl. Mech. Eng. 445 (2025) 118206.



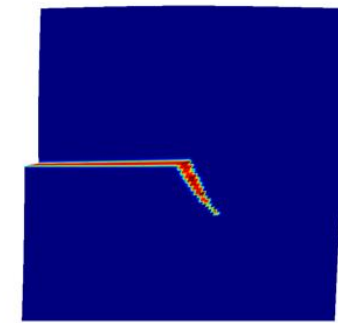
Singularity and mesh sensitivity eliminated



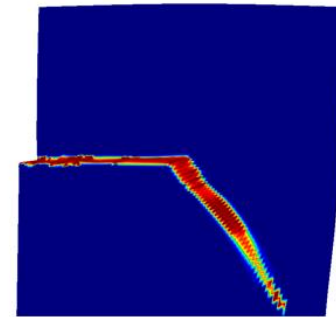
Benchmark test #2: shear test



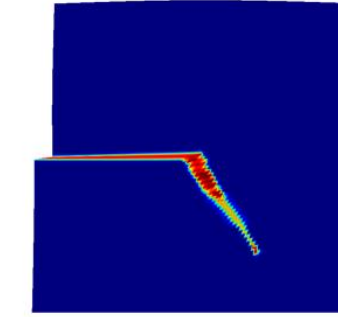
ux=0.6mm, Hencky-strain-based DHE model



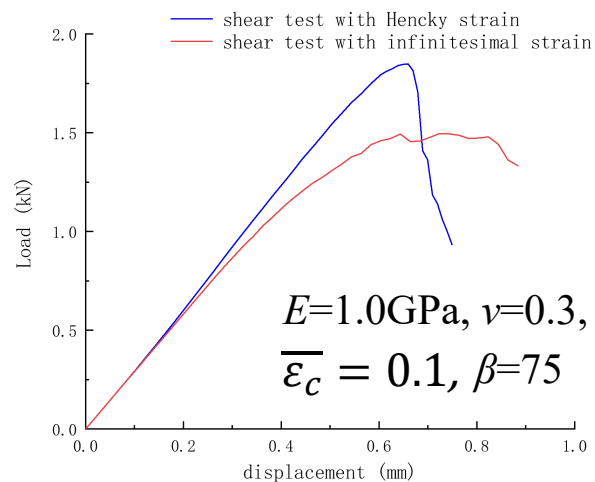
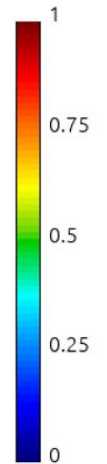
ux=0.6mm, infinitesimal-strain-based DHE model



ux=0.72mm, Hencky-strain-based DHE model

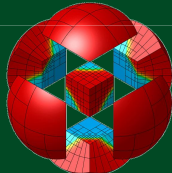


ux=0.72mm, infinitesimal-strain-based DHE model

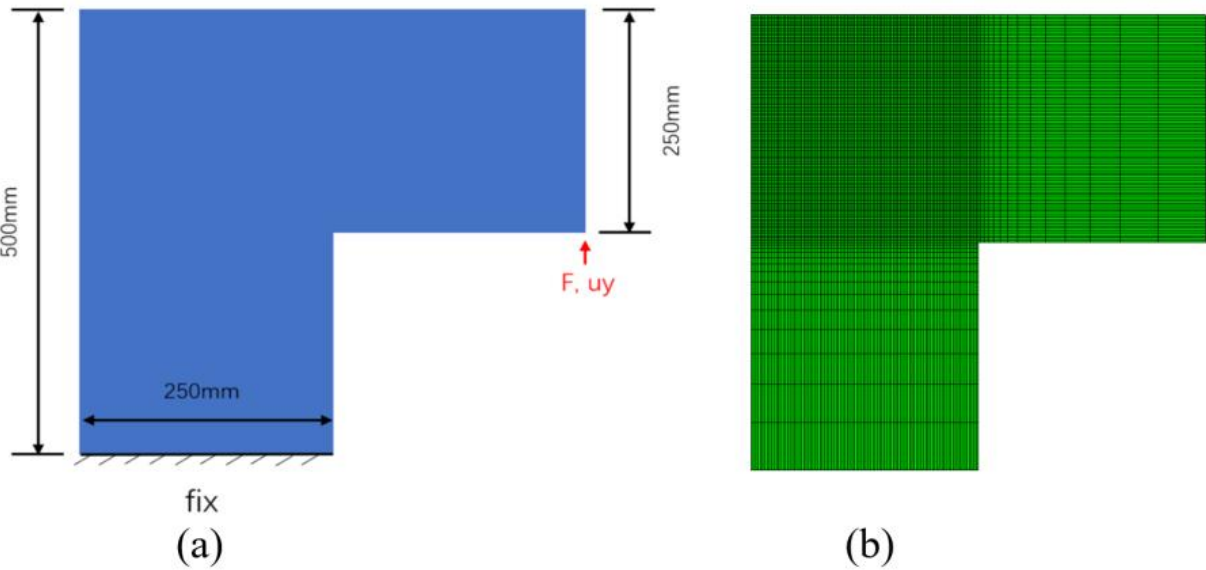


Finite-deformation kinematics affects the evolution of the principal tensile strain

Benchmark test #3: L-shaped concrete panel under mixed-mode fracture

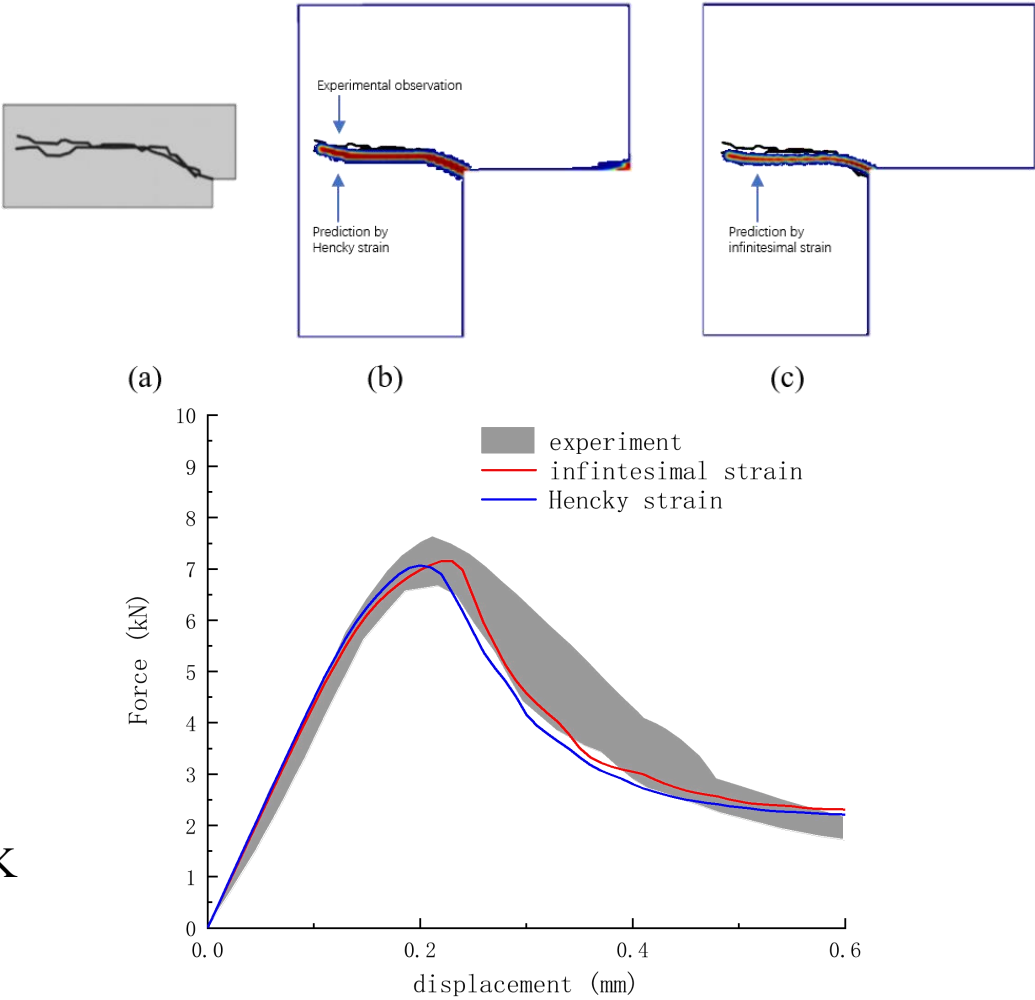


$E=18\text{GPa}$, $\nu=0.18$, $h=5.0\text{mm}$, $\overline{\epsilon}_c = 1.8 \times 10^{-4}$, $\beta=1765$



	Infinitesimal strain	Hencky strain
DOFs	258216	258216
Load increments	61	60
Converged increments	61	60
Average iterations for each increment	29	29
Max iterations	60	97
Physical time	1.14h	1.24h

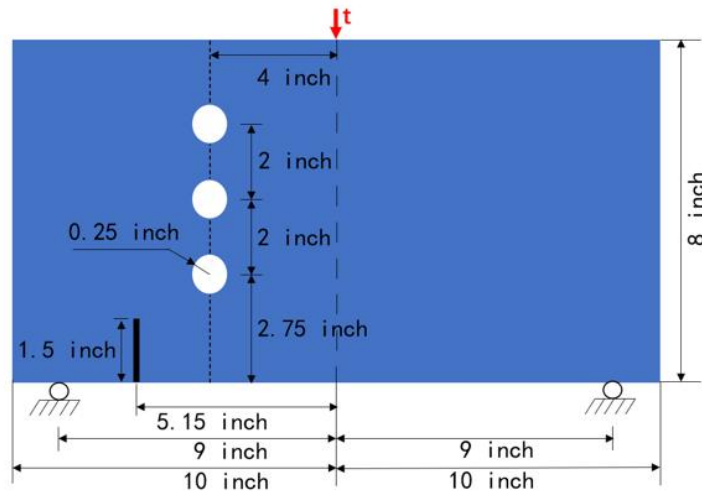
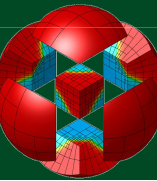
Intel® Core™ i7-13700K
with 12 cores



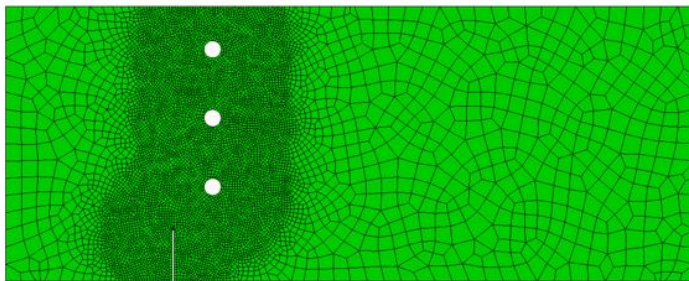
Cao Y.H., Zhang C.Y., Eng. Frac. Mech., 2026, submitted

Benchmark test #4: asymmetrically notched beam

MFEM Community
Workshop, 25/09/2026



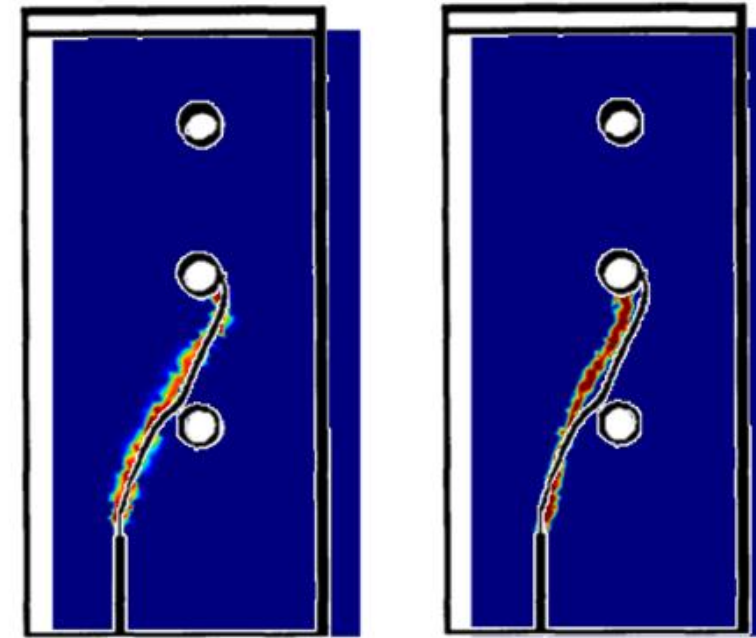
(a)



(b)

$E=4.75 \times 10^5 \text{ psi}$, $\nu=0.35$, $h=0.05 \text{ inch}$,

$$\overline{\varepsilon}_c = 0.005, \quad \beta=500$$



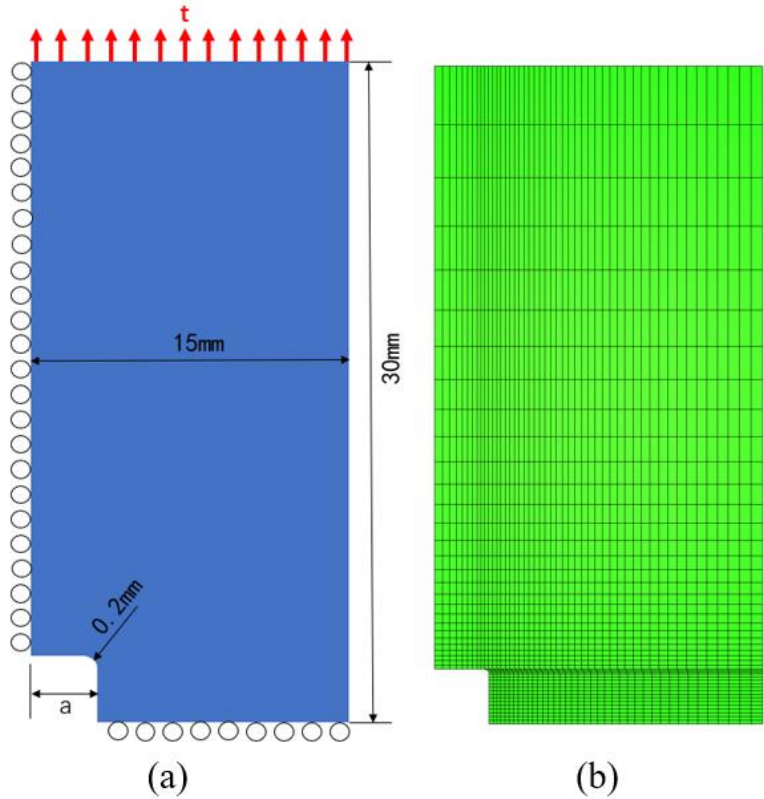
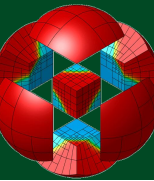
(a)

(b)

(a) Hencky strain (b) infinitesimal strain.

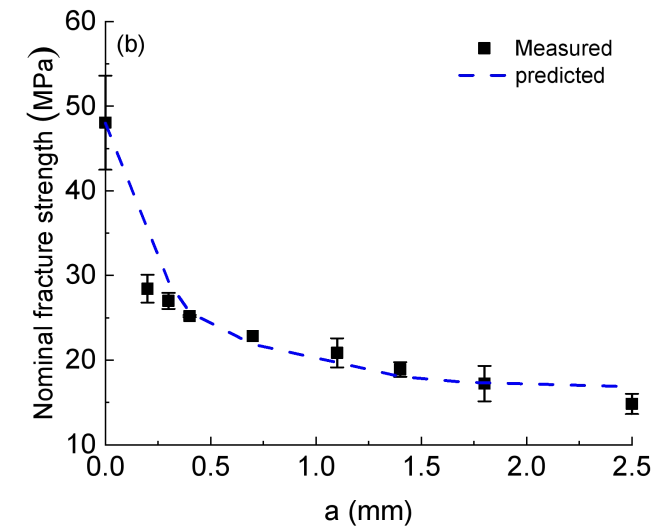
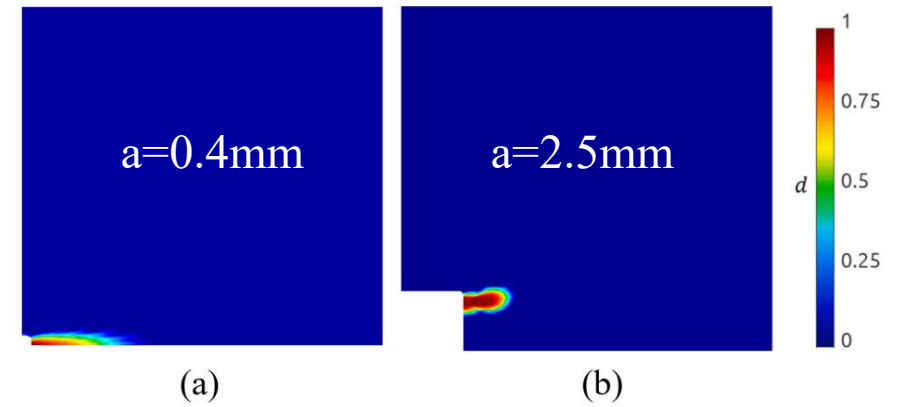
Cao Y.H., Zhang C.Y., Eng. Frac. Mech., 2026, submitted

Benchmark test #5: Scale-dependent fracture of specimens with square defects

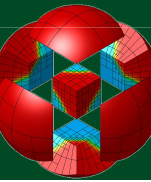


$$E=3.3\text{GPa}, \nu=0.35, h=1.1\text{mm},$$

$$\overline{\varepsilon}_c = 0.014, \quad \beta=650$$



Cao Y.H., Zhang C.Y., Int. J. Eng. Sci. 223 (2026) 104498.

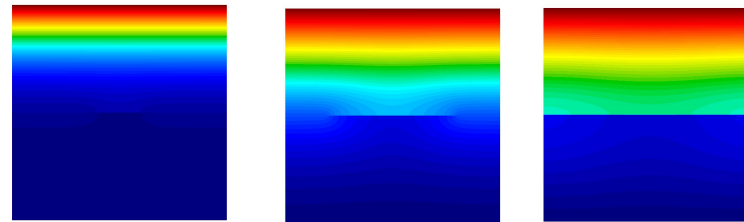


Major findings

1. Homogenization error: an important source of “nonlocal interaction”
2. Homogenized energy density: constitutive modeling of inhomogeneous deformation
3. FEM + ALM: accurate and robust for solution of high-order governing equations
4. High-order terms of strain energy density remove singularity of stress/strain

Future work

1. Thermal-damage coupling
2. Efficient iterative scheme for medium/large-scale problems
3. Higher-order model and solution method for ductile fracture





中山大學
SUN YAT-SEN UNIVERSITY

**Thanks for your attention.
Welcome to try or co-develop with us.**

`docker pull acculix/solvers:latest`



<https://github.com/chunyu79/HomogenizedEnergyTheory>

